COMPUTER SCIENCE - BS

The computer science curriculum is designed to prepare students to enter the rapidly expanding field of computing.

The four-year undergraduate curriculum in Computer Science at Texas A&M provides a sound preparation in computing, as well as in science, mathematics, English and statistics. Students take a broad set of core computer science courses in the early semesters, which exposes them to the main concepts in computing. During the later semesters, students take elective computer science courses drawn from three tracks (systems, software and languages, and information and applications) to provide both breadth and depth. The electives can be used to tailor the curriculum to match the student's interests. Graduate courses may be taken by qualified students for some of the electives.

A major in Computer Science includes complementary electives that allow students to design a course of study that takes advantage of opportunities offered by other departments across the university.

Program Mission

The mission of the computer science program is to prepare intellectual, professional and ethical graduates, capable of meeting challenges in the field of computer science.

Program Educational Objectives

The Program Educational Objectives of the Bachelor of Science in Computer Science program describe what the program's graduates are expected to attain within a few years of graduation:

1. Graduates continue using computer science principles to identify and solve emerging technological and societal problems, in an ethical and responsible manner.
2. Graduates who choose to enter the workforce will become effective leaders and innovators contributing through their influence on their teams, organizations and professional fields.
3. Graduates who choose to pursue advanced degrees will gain admission to and succeed in prestigious graduate programs.
4. Graduates continue engaging in life-long learning to adapt to new technologies, tools and methodologies needed to respond to a changing world.

This program is approved to be offered at the Texas A&M University at Galveston campus.

Program Requirements

The freshman year is identical for degrees in aerospace engineering, architectural engineering, biological and agricultural engineering, civil engineering, computer engineering, computer science, data engineering, electrical engineering, electronic systems engineering technology, environmental engineering, industrial distribution, industrial engineering, interdisciplinary engineering, manufacturing and mechanical engineering technology, mechanical engineering, multidisciplinary engineering technology, nuclear engineering, ocean engineering and petroleum engineering. The freshman year is slightly different for chemical engineering, biomedical engineering and materials science and engineering degrees in that students take CHEM 119 or CHEM 107/CHEM 117 and CHEM 120. It is recognized that many students will change the sequence and number of courses taken in any semester. Deviations from the prescribed course sequence, however, should be made with care to ensure that prerequisites for all courses are met.

First Year		
Fall		**Semester Credit Hours**
CHEM 107	General Chemistry for Engineering Students [1,4]	3
CHEM 117	General Chemistry for Engineering Students Laboratory [1,4]	1
ENGL 103 or ENGL 104	Introduction to Rhetoric and Composition [1] or Composition and Rhetoric	3
ENGR 102	Engineering Lab I - Computation [1]	2
MATH 151	Engineering Mathematics I [1,2]	4
University Core Curriculum (https://catalog.tamu.edu/undergraduate/general-information/university-core-curriculum/) [3]		3
	Semester Credit Hours	**16**
Spring		
ENGR 216/ PHYS 216	Experimental Physics and Engineering Lab II - Mechanics [1]	2
MATH 152	Engineering Mathematics II [1]	4
PHYS 206	Newtonian Mechanics for Engineering and Science [1]	3
University Core Curriculum (https://catalog.tamu.edu/undergraduate/general-information/university-core-curriculum/) [3]		3
Select one of the following:		3-4
CHEM 120	Fundamentals of Chemistry II [1,4]	
University Core Curriculum (https://catalog.tamu.edu/undergraduate/general-information/university-core-curriculum/) [3,5]		
	Semester Credit Hours	**15-16**
	Total Semester Credit Hours	**31-32**

1. A grade of C or better is required.
2. Entering students will be given a math placement exam. Test results will be used in selecting the appropriate starting course which may be at a higher or lower level.
3. Of the 21 hours shown as University Core Curriculum electives, 3 must be from creative arts (BS-AREN curriculum take ARCH 249, ARCH 250, or ARCH 350), 3 from social and behavioral sciences (BS-DAEN curriculum take ECON 202 or ECON 203; BS-IDIS curriculum take ECON 202), 3 from language, philosophy and culture (BS-AREN, BS-CVEN, BS-EVEN and BS-PETE curriculum take PHIL 482). In addition, 6 hours are required from American history and 6 hours from government/political science. The required 3 hours of international and cultural diversity and 3 hours of cultural discourse may be satisfied by courses that also meet the creative arts, social and behavioral sciences, language, philosophy and culture or American history requirements if those courses are on the approved list of international and cultural diversity (https://catalog.tamu.edu/undergraduate/general-information/degree-information/international-cultural-diversity-requirements/) courses and cultural discourse (https://catalog.tamu.edu/undergraduate/general-information/degree-information/cultural-discourse-requirements/) courses.

4 BS-BMEN, BS-CHEN and BS-MSEN require 8 hours of fundamentals of chemistry which are satisfied with CHEM 119 or CHEM 107/CHEM 117 and CHEM 120; Students with an interest in BS-BMEN, BS-CHEN and BS-MSEN can take CHEM 120 second semester freshman year. Students who take CHEM 119 and CHEM 120 can substitute CHEM 120 for CHEM 107/CHEM 117.

5 For BS-PETE, allocate 3 hours to core communications course (ENGL 210, COMM 203, COMM 205, or COMM 243) and/or 3 hours to University Core Curriculum elective. For BS-MEEN, allocate 3 hours to core communications course (ENGL 203, ENGL 210, COMM 203 or COMM 205) and/or 3 hours to University Core Curriculum elective. For BS-BAEN, allocate 3 hours to core communications course (ENGL 210)

		Semester Credit Hours
Second Year		
Fall		
CSCE 181	Introduction to Computing [1]	1
CSCE 120	Program Design and Concepts [1]	3
CSCE 222/ ECEN 222	Discrete Structures for Computing [1]	3
MATH 304	Linear Algebra [1]	3
University Core Curriculum (https://catalog.tamu.edu/undergraduate/general-information/university-core-curriculum/) [3]		3
Science elective [6]		4
	Semester Credit Hours	**17**
Spring		
CSCE 221	Data Structures and Algorithms [1]	4
CSCE 312	Computer Organization [1]	4
CSCE 314	Programming Languages [1]	3
Select one of the following:		3
COMM 203	Public Speaking	
COMM 205	Communication for Technical Professions	
ENGL 210	Technical and Professional Writing	
Complementary elective [1,7]		3
	Semester Credit Hours	**17**
Third Year		
Fall		
CSCE 231	Computing Ethics [1]	1
CSCE 313	Introduction to Computer Systems [1]	4
CSCE 331	Foundations of Software Engineering [1]	4
STAT 211	Principles of Statistics I [1]	3
Complementary elective [1,7]		3
	Semester Credit Hours	**15**
Spring		
CSCE 325	Foundations of Artificial Intelligence and Machine Learning [1]	3
CSCE 411	Design and Analysis of Algorithms [1]	3
CSCE 481	Seminar [1]	1
Select one of the following: [1]		3
MATH 251	Engineering Mathematics III	
MATH 308	Differential Equations	
STAT 212	Principles of Statistics II	
Computer science elective [1,8]		3
Science elective [6]		3
High Impact Experience [9]		0
CSCE 399	High-Impact Experience	
	Semester Credit Hours	**16**
Fourth Year		
Fall		
University Core Curriculum (https://catalog.tamu.edu/undergraduate/general-information/university-core-curriculum/) [3]		3
Complementary elective [1,7]		3
Computer science elective [1,8]		9
	Semester Credit Hours	**15**
Spring		
CSCE 482	Senior Capstone Design [1]	3
University Core Curriculum (https://catalog.tamu.edu/undergraduate/general-information/university-core-curriculum/) [3]		6
Computer science elective [1,8]		6
	Semester Credit Hours	**15**
	Total Semester Credit Hours	**95**

6 See advisor for list of acceptable science courses.

7 Complementary electives should be chosen only after consultation with a departmental advisor who will help the student arrange a program appropriate to his or her plans following graduation. Students will file a degree plan to get formal approval of complementary electives.

8 Of the 18 hours shown as computer science electives, 3 must be from systems electives, 3 must be from software and languages electives, 3 must be from information and applications electives, 9 must be from upper-level computer science and engineering electives. See advisor for list of acceptable course choices.

9 All students are required to complete a high-impact experience in order to graduate. The list of possible high-impact experiences is available in the Computer Science and Engineering advising office.

Total Program Hours 126